

PROJECT PLAN

Graph Learning for Interpretable Urban Accessibility and Opportunity

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Practicumgroep:

1

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1 Problem definition

Cities are spatially interconnected systems in which access to opportunity depends on public transport, street network structure and the distribution of destinations such as jobs and daily amenities. These factors do not exist independently. The accessibility of one place is partly determined by what can be reached from it and partly by the structure of its surrounding urban environment. This project studies that interaction across ten U.S. cities using multimodal urban data and graph based machine learning.

The project combines five data sources. GTFS provides public transport supply like routes, stops and service frequencies. OpenStreetMap (OSM) and OSMNX provide amenity locations and street network structure. LEHD/LODES provides workplace opportunity through spatially referenced job counts. TIGER/Line provides census block boundaries, which are used to define the spatial units and their adjacency relations. ACS provides socio demographic context that will be used after the modeling stage to analyze which groups are concentrated in high and low accessibility neighborhoods.

The main objective is to construct a block-level accessibility and opportunity score from reachable jobs and reachable amenities and then to test whether this score can be predicted from public-transport and built-environment characteristics. The target is defined through *reachable opportunities*, whereas the predictors mainly describe *transport supply* and *urban form*. Since neighboring blocks influence each other, the project compares both a strong non graph baseline and graph based models.

2 Keywords

Urban accessibility, opportunity, graph learning, GraphSAGE, graph convolutional networks, XGBoost, explainable AI, SHAP values, GNNExplainer, census blocks, GTFS, LEHD/LODES, OSMNX.

3 Literature review

The literature relevant to this project can be divided into three parts: accessibility measurement, machine learning for urban spatial analysis and graph-based explainable modeling.

First, accessibility research shows that accessibility is best understood through the opportunities that can be reached from a location, rather than through the local presence of infrastructure alone. Deboosere en El-Geneidy (2018) demonstrate this clearly in their study of job accessibility by public transport across eleven Canadian metropolitan areas. They treat workplace opportunity as a measurable spatial outcome and shows how such an outcome can be studied in a comparative multi city setting. Gong (2023) show that accessibility becomes more meaningful when it is derived from realistic spatial indicators rather than from simple geometric assumptions. Although their application concerns park access, the big idea is important here as well: accessibility should be constructed from what can be reached from a neighborhood.

Second, urban machine learning research shows that multimodal spatial analysis requires careful target construction and feature design. Casali, Aydin en Comes (2022) provide a review of machine learning for urban spatial analysis and show that studies in this area often combine transport, road-network, land-use and demographic data. Their review is useful because it shows both the potential of multimodal urban ML and the need for methodological clarity when defining targets and predictors. This is particularly relevant for the present project, where the accessibility target is constructed from reachable jobs and amenities, while the predictors are mainly public transport and built environment indicators.

Third, graph-based learning provides a natural framework for modeling spatial dependence between urban areas. Xie e.a. (2025) offer the closest methodological reference for this project. They represent urban blocks as graph nodes, use graph neural networks to predict a block-level urban outcome, and apply explainable AI to understand the local drivers of the predictions. Their work shows that graph learning can capture neighborhood effects more explicitly than standard tabular models and that local explanations are necessary if the results are to be analytically useful. For the algorithms themselves, Hamilton, Ying en Leskovec (2017) provide the GraphSAGE framework, Kipf en Welling (2017) provide the graph convolutional network (GCN), Chen en Guestrin (2016) provide the main non-graph baseline XGBoost, Lundberg en Lee (2017) provide SHAP for tabular explainability, and Ying e.a. (2019) provide GNNExplainer for local explanations of graph neural networks.

Taken together, accessibility can be represented through reachable opportunities, urban ML can integrate heterogeneous spatial datasets and graph learning can model the spatial dependence between neighborhoods while remaining interpretable through SHAP and GNNExplainer.

4 Research Questions

There are two research questions proposed in this thesis:

RQ 1: To what extent can XGBoost, GraphSAGE and GCN predict a block level accessibility and opportunity score from public-transport supply and built-environment indicators across ten U.S. cities?

RQ 2: Which spatial factors contribute most strongly to accessibility and opportunity and how can local explanation methods and clustering reveal recurring patterns such as transport-poor or opportunity-poor neighborhoods?

5 Method and Approach

The project can be divided into five stages:

1. data collection and harmonization,
2. target construction,
3. graph construction and predictive modeling,
4. explainability and clustering,
5. socio-demographic interpretation and visualization.

5.1 Study area and spatial unit

The study covers ten U.S. cities: New York, Boston, Washington DC, Atlanta, Houston, Denver, Tulsa, Little Rock, Santa Cruz and Los Angeles. The main spatial unit is the **census block**. Each block becomes one observation in the tabular dataset and one node in the graph. TIGER/Line provides the block geometry and identifiers, from which block centroids are computed.

The choice for census blocks follows the aim of keeping the analysis spatially detailed. Blocks are small enough to reflect local transport and urban form differences, yet structured enough to be linked to GTFS stops, OSM amenities, LEHD job counts and block adjacency. If severe sparsity or runtime problems arise, census tracts will be the fallback unit, but the main design is based on blocks.

5.2 Data collection and preprocessing

The following data sources are collected for each city:

- **GTFS:** public-transport schedules, stops, routes and service frequencies;
- **OSM/OSMNx:** street-network geometry and amenity locations;
- **LEHD/LODES:** workplace locations and counts;
- **TIGER/Line:** census-block polygons and adjacency;
- **ACS:** socio-demographic context variables.

The raw files are not used directly as model input. Instead, they are transformed into block-level variables and network-based accessibility measures. All spatial layers are projected to a common coordinate reference system. For each block, its centroid is used as the origin point for opportunity reachability calculations. OSM street networks and GTFS stops are spatially linked to blocks. LEHD job counts are assigned to destination blocks and later queried within distance bands around origin blocks. ACS variables are processed separately and attached only in the final interpretation stage.

5.3 Target construction: accessibility and opportunity score

The supervised target is a continuous **block-level accessibility and opportunity score**. The score is constructed from two components:

- employment accessibility, based on reachable jobs;
- amenity accessibility, based on reachable amenities.

This target design is the key methodological choice of the project. It makes the learning problem non-circular: the target is based on *opportunities that can be reached*, whereas the predictors mainly describe *public-transport supply* and *built-environment structure*.

5.3.1 Employment accessibility

Let b denote an origin block and j a destination block containing jobs. For a radius r , the number of reachable jobs from block b is defined as

$$J_b(r) = \sum_{j \in \mathcal{B}} Jobs_j \cdot \mathbf{1}(d^{net}(b, j) \leq r),$$

where

- $\mathcal{B}$ is the set of all destination blocks in the city,
- $Jobs_j$ is the number of jobs in destination block j ,

- $d^{net}(b, j)$ is the shortest-path distance on the street network between the centroids of blocks b and j ,
- $\mathbf{1}(\cdot)$ is the indicator function.

This is computed for

$$r \in \{1, 2, 3, 4, 5\} \text{ km.}$$

This yields five employment-accessibility values for every block.

5.3.2 Amenity accessibility

Amenity accessibility is defined analogously. Let $\mathcal{A}_c$ be the set of amenities of type c , where c belongs to a fixed set of categories such as food/retail, healthcare, education, recreation, and public services. Then the number of reachable amenities of type c from block b is

$$M_{b,c}(r) = \sum_{a \in \mathcal{A}_c} \mathbf{1}(d^{net}(b, a) \leq r),$$

where $d^{net}(b, a)$ is the shortest-path distance from the centroid of block b to amenity a .

Because some amenity categories are much more common than others, counts are first normalized within category. The amenity-accessibility component for radius r is then defined as the mean over categories:

$$M_b(r) = \frac{1}{|C|} \sum_{c \in C} \widetilde{M}_{b,c}(r).$$

Again, this yields five values per block.

5.3.3 Normalization

The raw job and amenity counts differ strongly in scale, so all accessibility components are normalized city by city using min-max scaling:

$$z_{bk} = \frac{x_{bk} - \min(x_k)}{\max(x_k) - \min(x_k)}.$$

This means the final target is interpreted as a *relative within-city accessibility and opportunity score*. The main analytical focus is therefore on spatial structure within each city and on general patterns across cities, rather than on direct absolute comparisons of raw opportunity volume between very different urban systems.

5.3.4 Distance aggregation

Closer opportunities should contribute more strongly than distant ones. Therefore, a distance-decay weighting is applied:

$$\alpha_1 = 0.30, \quad \alpha_2 = 0.25, \quad \alpha_3 = 0.20, \quad \alpha_4 = 0.15, \quad \alpha_5 = 0.10.$$

The aggregated employment accessibility of block b is

$$E_b = \sum_{k=1}^5 \alpha_k \widetilde{J}_b(r_k),$$

and the aggregated amenity accessibility is

$$A_b = \sum_{k=1}^5 \alpha_k \widetilde{M}_b(r_k).$$

5.3.5 Final target score

The final target score is

$$Y_b = \lambda E_b + (1 - \lambda) A_b.$$

The main specification uses equal importance for employment and amenities:

$$Y_b = 0.5 E_b + 0.5 A_b.$$

Two alternative specifications are used as robustness checks:

$$Y_b^{(1)} = 0.6 E_b + 0.4 A_b, \quad Y_b^{(2)} = 0.4 E_b + 0.6 A_b.$$

In addition, a sensitivity analysis will test whether conclusions remain stable under alternative radius choices and under a reduced amenity category set. This is important because the target contains design choices that are reasonable but not uniquely determined.

5.4 Predictor construction

The predictor set contains **public-transport** and **built-environment** variables. ACS variables are deliberately excluded from the core predictive model and are used only after the modeling stage for equity interpretation.

Public-transport predictors. From GTFS, the following block-level predictors are constructed:

- stop or station count,
- stop density,
- total weekday service frequency,
- number of distinct routes,
- modal variety,
- average scheduled headway.

Built-environment predictors. From OSM/OSMNX and TIGER/Line, the following block-level predictors are constructed:

- total street length density,
- bikeable street length density,
- intersection density,
- road-network connectivity,
- block compactness,
- land-use diversity.

Raw amenity counts are not reused as predictors, because they already contribute to the target through A_b . This avoids leakage. Likewise, the exact job-access variables used in the target are not reused unchanged as predictors.

The final feature vector for block b is therefore

$$\mathbf{x}_b = [x_{b1}^{PT}, \dots, x_{bp}^{PT}, x_{b1}^{BE}, \dots, x_{bq}^{BE}].$$

5.5 Graph construction

The urban area is represented as an adjacency-only graph

$$G = (V, E),$$

where every census block is a node:

$$V = \{b_1, b_2, \dots, b_n\},$$

and an edge is added between two blocks if their polygons share a boundary:

$$E = \{(b_i, b_j) \mid b_i \text{ and } b_j \text{ are adjacent}\}.$$

This graph is intentionally simple. It is spatially interpretable, feasible within the thesis timeframe, and sufficient to test whether spatial context improves prediction. Since the study covers ten cities, the combined graph is a disconnected graph with one connected component per city.

5.6 Predictive models

Three regression models are used to predict Y_b .

XGBoost. The first model is XGBoost, used as a strong non-graph baseline. For a dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$, the prediction is

$$\hat{y}_i = \sum_{t=1}^T f_t(\mathbf{x}_i),$$

where each f_t is a regression tree. The objective function is

$$\mathcal{L} = \sum_{i=1}^N l(y_i, \hat{y}_i) + \sum_{t=1}^T \Omega(f_t),$$

with l a regression loss and Ω a regularization term.

GraphSAGE. The second model is GraphSAGE. For node v at layer k , the update is

$$h_v^{(k)} = \sigma\left(W^{(k)} \cdot \text{AGG}\left(\{h_v^{(k-1)}\} \cup \{h_u^{(k-1)} : u \in \mathcal{N}(v)\}\right)\right),$$

where $h_v^{(0)} = \mathbf{x}_v$.

GCN. The third model is a graph convolutional network. A standard layer update is

$$H^{(k+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(k)} W^{(k)}\right),$$

where $\tilde{A} = A + I$ is the adjacency matrix with self-loops, $\tilde{D}$ is the corresponding degree matrix, $H^{(0)} = X$ and $W^{(k)}$ are trainable parameters.

All three models operate in a regression setting and use Y_b as the target.

5.7 Explainability and clustering

To interpret the models:

- XGBoost is explained with SHAP values;
- GraphSAGE and GCN are explained with GNNExplainer.

SHAP provides global and local feature contributions for the tabular baseline. GNNExplainer provides local explanations for the graph models by identifying which node features and neighboring nodes are most influential for a prediction.

The resulting explanation vectors are then clustered, for example with k -means, to identify recurring spatial profiles. The aim is to move from single-block explanations to interpretable neighborhood types, such as transport-poor blocks, street-network-limited blocks or blocks with strong opportunities despite modest transport supply.

5.8 ACS-based interpretation

After the main modeling stage, ACS variables are linked to the spatial units and used for socio-demographic interpretation. Because ACS variables are not available at true census-block level, they will be attached through the most appropriate higher-resolution geography, such as block groups or tracts.

The ACS stage addresses questions such as:

- which groups are concentrated in the lowest accessibility deciles,
- whether low-accessibility areas are associated with income or demographic patterns,
- whether different explanation clusters correspond to different social contexts.

6 Evaluation

Models are evaluated with the standard regression metrics

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2},$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|,$$

and

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}.$$

A simple random node split would create spatial leakage, especially for the graph models. Therefore, the main evaluation will be **spatially aware**. Within each city, blocks will be partitioned into spatial clusters or contiguous subareas and holdout sets will be defined at that level rather than by randomly sampling individual blocks. In addition, a leave-one-city-out experiment will be used as a transferability test across urban environments.

Evaluation is not limited to predictive accuracy. The explanation stage is also evaluated qualitatively:

- do SHAP and GNNExplainer identify plausible spatial drivers,
- do clustered explanation profiles correspond to interpretable neighborhood types,
- does the ACS-based interpretation reveal clear socio-demographic inequalities?

The project pipeline can therefore be summarized as follows:

- **Data preprocessing**
 - Collecting GTFS, OSM/OSMNX, LEHD/LODES, ACS and TIGER/Line data.
 - Harmonizing all sources at census-block level.
- **Target construction**
 - Computing reachable jobs within 1–5 km.
 - Computing reachable amenities within 1–5 km.
 - Normalizing and aggregating these into Y_b .
- **Feature construction**
 - Constructing PT predictors from GTFS.
 - Constructing BE predictors from OSM/OSMNX and TIGER/Line.

- **Graph construction**

- Representing census blocks as nodes.
- Connecting adjacent blocks with edges.

- **Prediction**

- Training XGBoost, GraphSAGE and GCN to predict Y_b .

- **Explanation and clustering**

- Applying SHAP and GNNExplainer.
- Clustering explanation profiles.

- **Interpretation and visualization**

- Using ACS for post hoc socio-demographic interpretation.
- Building the prototype dashboard.

7 Plan

The project is divided into four phases: exploration, implementation, evaluation, and finalizing. Writing will take place throughout the project, so that methodological choices and intermediate results are documented while the research is still in progress.

A milestone is placed after week 6. By then, the harmonized dataset, the final accessibility target, and the first working model implementations should be available. If the project falls behind schedule, the priority is to preserve the core contribution: target construction for all cities, XGBoost and at least one graph model, one explanation method per model family, and a lightweight dashboard.

Week	Activity
1	Literature review and familiarizing with the datasets
2	Finalizing the project plan and fixing the target and feature definitions
3	Collecting and harmonizing GTFS, OSM/OSMNX, LEHD/LODES, ACS, and TIGER/Line data
4	Computing reachable jobs and reachable amenities for all cities
5	Constructing the final accessibility/opportunity target and building the city graphs
6	Implementing XGBoost, GraphSAGE, and GCN; writing the first thesis draft sections
7	Training and validating the predictive models; preparing the midterm presentation
8	Applying SHAP and GNNExplainer
9	Clustering explanation profiles and linking ACS variables for interpretation
10	Building the prototype dashboard and consolidating results
11	Finalizing thesis writing and revising figures and analysis
12	Finalizing thesis writing and preparing the presentation

Tabel 1: Weekly thesis plan

Referenties

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A Bijlage L^AT_EX code

Bijgevoegd zijn de [code](#) en [bibliografie](#).
